A COMPANION TO THE AMERICAN SUPPLEMENT INVESTIGATION

The Supplement Primer

What the label doesn't tell you, in plain English



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Based on *The American Supplement Investigation*, a three-book series: *What the Label Doesn't Tell You*, *The Hollow Aisle*, and *Prove It*, by Robert Coleman, MS Pharm., with Joy Meier, Pharm.D., and Julio Lopez, Pharm.D.

© 2026 Robert Coleman. Published by Coleman Publishing, part of Broken Promises Healthcare. The full investigation, with sources, is free to read at supplements.brokenpromiseshealthcare.org.

Please read. This book is general education, not medical advice. It does not tell you what to take or avoid, and it does not suggest doses. Talk to your doctor or pharmacist about your own health, your medicines, and any supplement you are thinking of using. Product names that appear in the larger investigation are discussed there, with sources.

Contents

Welcome 4

PART 1: THE BIG PICTURE

1 What exactly is a supplement? 5

2 Two roads to the shelf 7

3 Who checks after it's on the shelf? 9

PART 2: LABELS AND CLAIMS

4 Reading a label, zone by zone 11

5 Tricks of the numbers 13

6 Claims decoded 15

7 How the sale works 17

PART 3: THINKING ABOUT EVIDENCE

8 Why "it worked for me" fools us 19

9 The evidence ladder 21

10 A cautionary tale: beta-carotene 23

11 What a good trial looks like 25

PART 4: SAFETY

12 "Natural" does not mean safe 27

13 The dose makes the poison 29

14 Supplements and your medicines 31

PART 5: WHAT THE EVIDENCE SHOWS

15 The report card 33

16 Do I need one? Three zones 35

PART 6: YOUR TURN

17 Your five-question checklist 37

BACK OF THE BOOK

Plain-English glossary 39

Where these facts come from 40

Your next two stops 41

Finding your way. Tap any line above to jump there. The edge tabs mark the six parts; the lit tab shows where you are. Tap ← **Contents** at the top of any page to return.

Read a label 11 · Judge a claim 15 · Spot a good trial 25 · Medicine interactions 31 · The checklist 37

Welcome



About three in four American adults take a dietary supplement. Many of them assume that something sitting on a store shelf next to the aspirin has been checked by the government for safety and for whether it works.

For supplements, that is usually not true. This short primer explains, in plain language and with pictures, what the law does and does not require, how to read a label, how to judge a health claim, what the best evidence shows, and what to ask before you buy.

Three promises

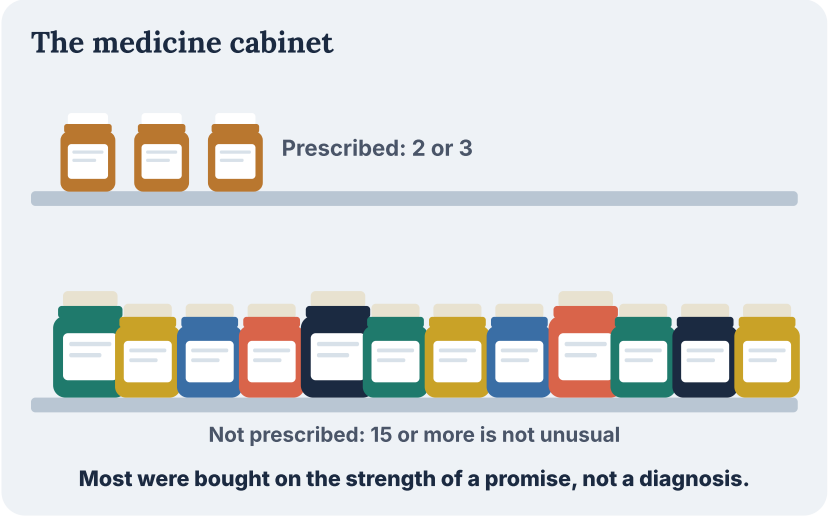
- **No hype in either direction.** Some supplements help some people. Many do not do what their labels suggest. We show you how to tell the difference.
- **No product advice.** We do not tell you what to take. We help you ask better questions.
- **Every fact has a source.** This primer is drawn from a three-book investigation with full citations.



How to use this book
Read it in order, or jump to a chapter. Each chapter is two pages and ends with a short bottom line. The five-question checklist on page 37 is the one page to keep.

1 What exactly is a supplement?

Open a typical bathroom cabinet and you will find two or three prescription bottles. Open the kitchen cupboard, the shelf by the coffee maker, or the gym bag by the door, and you will find a lot more: a multivitamin, fish oil, vitamin D, something for joints, something for sleep, a greens powder. None were prescribed. Most were bought because of a promise, not a diagnosis.



Wanting to stay healthy, feel better, or hedge against a thin diet is a perfectly reasonable goal. The question this book asks is simpler: is the product in your hand the right tool for it, and how would you know?

Here is the surprise

Under U.S. law, a dietary supplement is not a drug. It is a category of **food**. That one legal decision explains almost everything else in this book.

3 in 4
U.S. adults take a supplement of some kind

$60 billion
spent each year in the United States

$200 billion
global market every year

This is not a niche habit. It is close to a national default. When Congress passed the law that still governs supplements in 1994, there were about 4,000 products on the market. Today there are more than 80,000.

Supplement products on the U.S. market

Each bottle = 4,000 products

1994 (when the law passed)

about 4,000

Today more than 80,000



Twenty times as many products. The rules did not grow with them.

Many people take supplements to prevent disease, to feel more energetic, to age well, to boost immunity, or as insurance in case their diet falls short. Those are understandable reasons. They are also reasons that marketing is very good at finding.

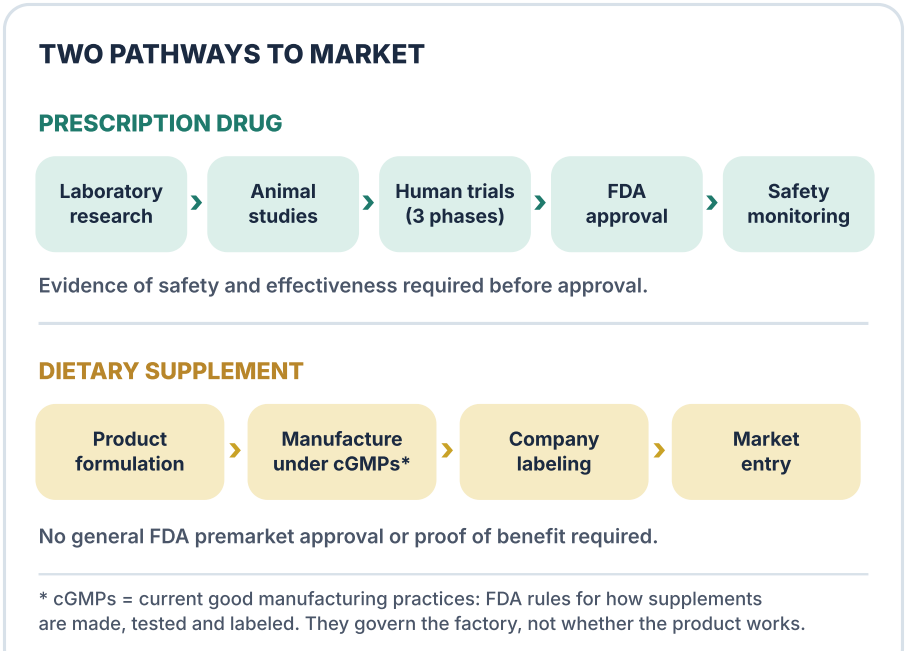
Bottom line

"Supplement" is a legal category, not a guarantee. It tells you how a product is sold, not whether it works.

Two roads to the shelf

2

Before a new prescription drug can be sold, its maker must take it through years of lab work, animal tests, and three phases of human trials. The U.S. Food and Drug Administration (FDA) reviews all of it before the drug can be sold at all. The law starts from the assumption that a drug is unproven and possibly unsafe until its maker shows otherwise.



A supplement follows almost none of that. Under a 1994 law, the Dietary Supplement Health and Education Act (DSHEA), a company can make a new supplement and sell it without showing anyone evidence that it works, and usually without even telling the FDA it exists. The law starts from the opposite assumption: a supplement is treated as safe until the FDA can show, after the fact, that it is not.

What the law does require

- **Current good manufacturing practices (cGMPs)** for identity, purity and consistency, enforced by inspections.
- **No disease claims.** A label cannot say a supplement treats, cures or prevents a disease.
- **A heads-up for new ingredients.** An ingredient not sold before October 1994 needs notice to the FDA with some evidence of safety, far less than a drug requires.

What it does not require

Proof that the product does what its marketing suggests. A claim like "supports heart health" needs no FDA review or approval before it is used, only a small-print disclaimer. The company is expected to have some support for it, but nobody checks first.



A telling example

Red yeast rice, sold as a supplement for "cholesterol support," can contain, depending on the batch, a compound chemically identical to the active ingredient in a prescription statin. One product was tested for years. The other, usually, was not. The shelf does not tell you which is which.



Bottom line

The difference between a drug and a supplement is not mainly what is inside. It is what the law required someone to prove before you could buy it.

Who checks after it's on the shelf?

3

More than 80,000 products. Thousands of stores and websites. Tens of millions of buyers. And no review before sale. So what happens if one turns out to be dangerous?

Almost never an inspection. The system mostly runs on **after-the-fact reports**. A person or doctor notices a possible reaction and files a report with the FDA, through a system called CAERS (the FDA's Adverse Event Reporting System for foods and supplements). Between 2004 and 2021 the FDA received more than 79,000 such reports tied to supplements. One widely cited review of 2004 to 2013 found more than 15,000 reports that included 339 deaths and nearly 4,000 hospitalizations.

From a harm to an action: every arrow can break



79,000+ adverse event reports tied to supplements, 2004 to 2021

Experts call that an undercount: most harms are never reported.

Surveillance happens after people have already been exposed.

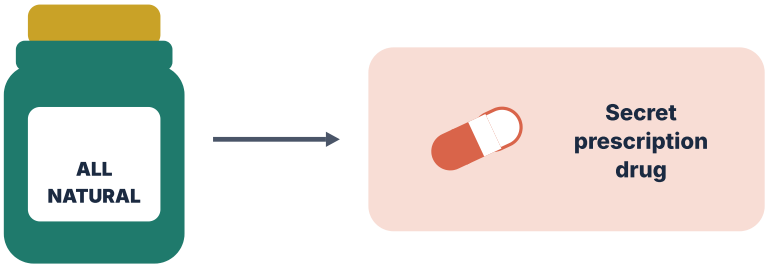
Experts who study the system say these numbers are an undercount. Most bad reactions are never reported, and many that are reported are never traced to a specific product.

What the FDA can do

It can send a warning letter, publish a public alert, or go to court to stop a clearly dangerous product. Its power to order a supplement recall directly is narrow and rarely used. Most supplement recalls are technically voluntary, negotiated after the problem is documented.

Hidden drugs

The FDA keeps a public list of supplements, mostly sold for weight loss, sexual performance or bodybuilding, found to contain secret prescription drugs, sometimes at doses higher than a doctor would ever prescribe. Many were sold as "all natural."



What the label says is not always what is inside.

Products made overseas or sold through online marketplaces are harder still to trace back to anyone responsible.

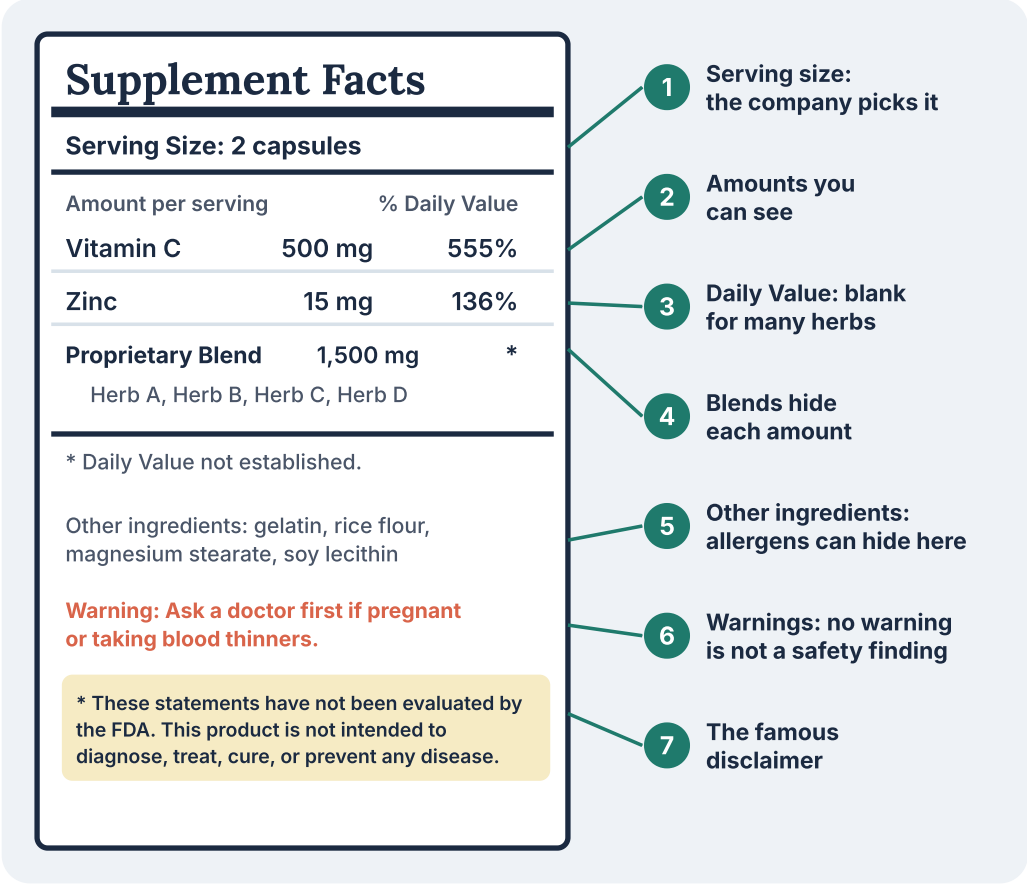
Bottom line

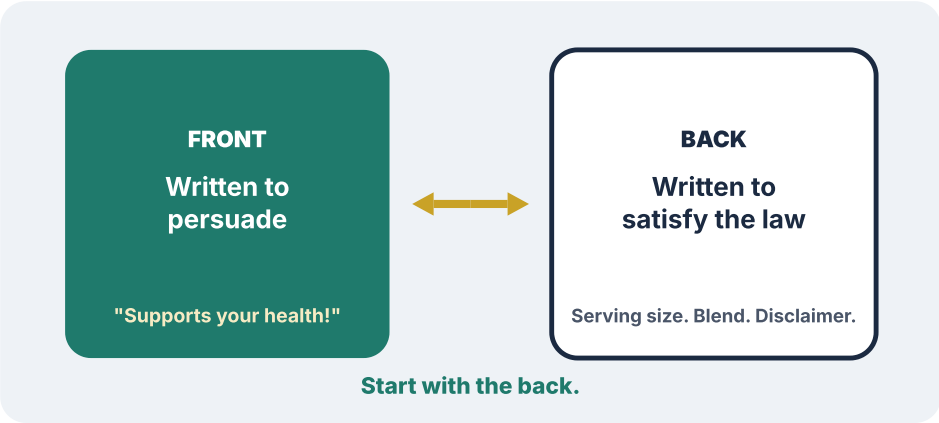
Safety checks mostly happen after people have already used a product. Treat a lack of recalls as a gap in the system, not as proof of safety.

Reading a label, zone by zone

4

The "Supplement Facts" panel looks a lot like the "Nutrition Facts" label on a cereal box: same tidy grid, same reassuring look. That resemblance does a lot of persuading. The two labels follow very different rules. (Amounts on the panel are given in milligrams, abbreviated mg: thousandths of a gram.)





Read in this order

1. **The disclaimer.** "These statements have not been evaluated by the Food and Drug Administration." It tells you the claim was not reviewed before sale, and it is the sentence people read least.
2. **Any "proprietary blend."** If amounts are hidden, you cannot judge the dose.
3. **The serving size.** The company chooses it. Check how many capsules make a serving.
4. **Only then, the claim on the front.** Regulations first, marketing last.

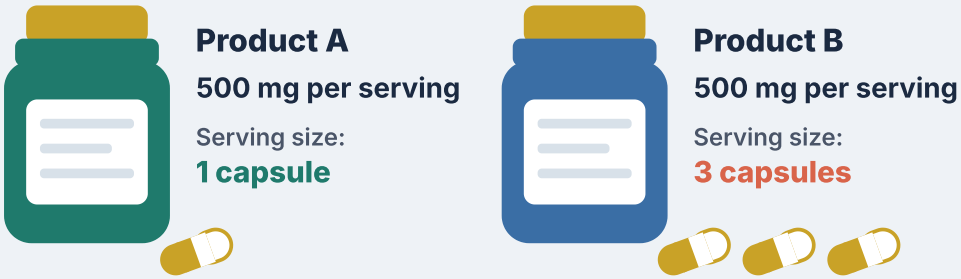
Why this order works
The front of the bottle is written to persuade. The back is written to satisfy the law. Reading the back first lets you see what is actually promised, and what is not.

Tricks of the numbers

5

Every number on the panel is "per serving," and the company decides what a serving is. A product can look strong by calling two or three capsules one serving. That is legal. It is also worth checking before you compare prices or strength.

Same big number, different serving



Product B needs three times as many capsules for the same 500 mg.

Percent Daily Value works well for vitamins like C or calcium, which have official daily values. For many herbs and newer compounds, no such standard exists, so the column is blank, right where you might need the most help interpreting the number.

The proprietary blend

Many sports, energy and "focus" products list a **proprietary blend**. The label gives the total weight of the blend, but keeps each ingredient's amount secret.

One blend, two very different products

The label says: Proprietary Blend, 1,500 mg total

Possibility 1



Possibility 2



The label is legal both ways. You cannot tell which one you are buying.

A label can legally tell you that a blend contains 1,500 mg of several ingredients without telling you whether the ingredient that drives the marketed effect makes up 1,400 mg or 100 mg.

Small print, big deal

The "other ingredients" list names fillers, binders, coatings and flavorings. If you have an allergy, read it. A missing warning does not mean a product is safe. It may mean nobody has looked closely enough yet.

Claims decoded

6

"Supports healthy blood sugar already within a normal range." Read that quickly in a store aisle and it sounds like help for diabetes. Read it slowly, the way a lawyer does, and it promises almost nothing: *support*, not treatment; *healthy*, not diabetic; *already normal*, not high.

What you hear and what it legally means

	WHAT YOU HEAR	WHAT IT LEGALLY MEANS
Supports heart health	Fewer heart attacks	Plays some role in normal heart function
Supports immune function	Fewer, milder colds	Involved in some part of immune activity
Promotes healthy aging	Slows aging	Nothing specific enough to test
Supports cognitive function	Better memory	Relates to normal brain function
Maintains healthy blood sugar	Helps diabetes	Only for levels already normal

None is a lie in the narrow legal sense. Each one invites you to hear more.

Two kinds of health claims

Disease claim
Says a product diagnoses, treats, cures or prevents a disease. This turns a product into an unapproved drug and can trigger enforcement.

Structure/function claim
Says a product "supports" a normal body function. Needs **no** FDA approval and no proof submitted first, only the small-print disclaimer.

Much of supplement marketing is the art of getting as close to the disease claim as the law allows without crossing it. The words, the product name, the photos and the testimonials then carry the rest of the message.



The gap is the marketing

No law requires it to be disclosed anywhere on the package.

Try this
Take any supplement ad and ask: what does the sentence actually promise, and what does my mind fill in? The gap between the two is the marketing.

How the sale works

7

Marketers describe the path from a stranger's attention to a sale in five steps. Knowing them helps you spot where you are in the process.

The five-step path from worry to sale



Belief becomes the next ad's testimonial, and the loop starts again.

- **Concern.** A worry about energy, aging, weight or sleep is usually already there.
- **Advertisement.** It joins your worry to a fix in one frame.
- **Promise.** The claim, stated or implied.
- **Purchase.** Often sped up by a countdown or a limited-time price.
- **Belief.** A happy customer becomes the next ad's testimonial.

Notice the loop
Next time an ad grabs you, ask which of the five steps you are in. Naming the step takes away some of its pull.

Words that sound like evidence



"Doctor recommended"
Can mean one physician, paid or not, said so on camera. It does not mean a medical body reviewed the product.



"Clinically studied"
Can mean one tiny, unpublished, company-funded study or a large independent trial. The label does not say which.



"Natural"
Describes where an ingredient came from, not how safe it is, how much is in it, or what it does.



"Ancient remedy"
Tells you something has been used for a long time. It says nothing about whether it works or what the risks are.

Influencers borrow trust. A creator you have followed for years may be paid whether or not the product works for you, often through a link that pays a share of your purchase. Paid endorsements are legal when clearly disclosed. It is worth looking for the disclosure.

Why "it worked for me" fools us

8

A woman struggled to sleep for years. A coworker suggested magnesium. Within ten days she slept better, and she has taken it every night since. Ask her if it works and she will say yes without hesitating.

There is no reason to doubt that she sleeps better. The real question is narrower: was it the magnesium that changed? Stories are powerful because people are built to learn from narrative. "It worked for me" beats a table of numbers almost every time, even for people who know better.

Four things besides the pill that can explain "it worked"



Confirmation bias

You notice the good days and explain away the bad



Placebo effect

Expecting to feel better can make you feel better



Regression to the mean

We start at our worst; bad patches often fade alone



Natural ups and downs

Symptoms come and go with or without a supplement

Four explanations to rule out first

- **Confirmation bias.** Once you believe it helps, you notice the good days and explain away the bad ones.
- **The placebo effect.** Real and often strong, especially for tiredness, mood, pain and sleep. Expecting to feel better can make you feel better.
- **Regression to the mean.** We try things at our worst moment, and a bad stretch tends to improve on its own.
- **Natural ups and downs.** Many symptoms come and go no matter what you take.

None of this requires anyone to be foolish or dishonest. It is how every human brain works, including mine.



Bottom line

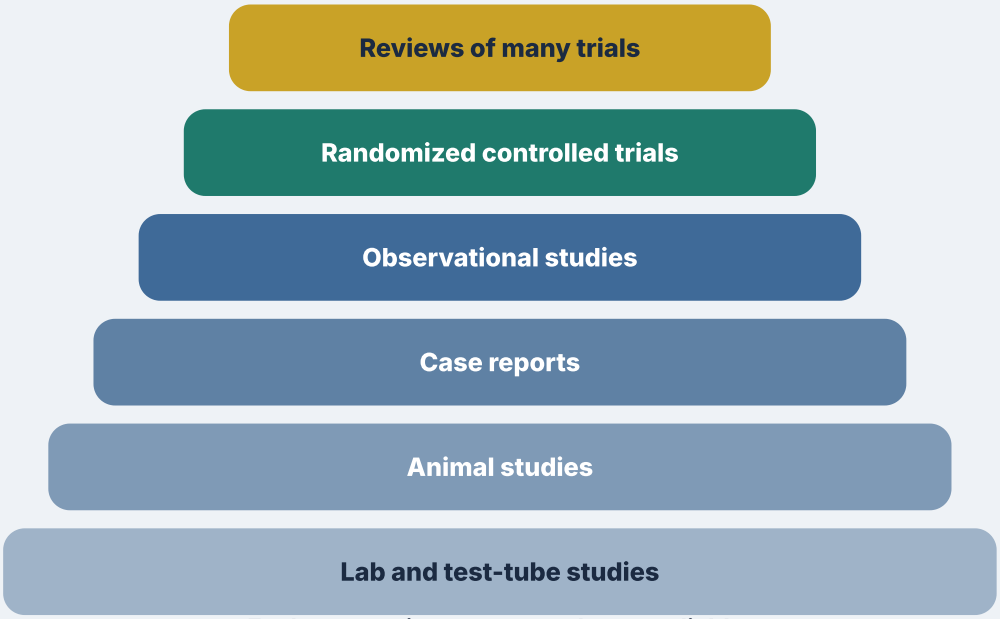
Personal stories are good for raising questions and poor for answering them. They can suggest an idea is worth testing. They cannot show that it works.

The evidence ladder

9

"Scientifically proven" is one of the most common claims in health marketing and one of the least examined. Proven how? By whom? Not all evidence ranks the same, and you can picture the kinds as steps on a ladder.

The evidence ladder, from weakest to strongest



Each step up is stronger and more reliable.

Test-tube and animal studies are useful for finding possible mechanisms but are far from what happens in a person. Observational studies follow real people but cannot rule out other explanations. A **randomized controlled trial** assigns people by chance to a treatment or a placebo, which is why it can show cause and effect.

Two traps to watch for

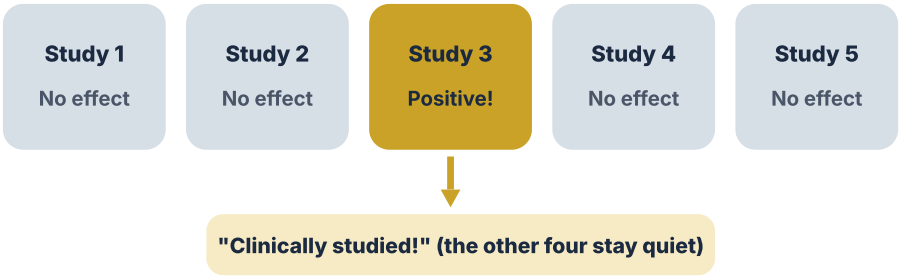
Significant is not the same as important

A result can be "statistically significant," meaning unlikely to be chance, and still be too small to matter to how you feel. Marketing leads with the first and leaves out the second.

Good news travels faster

Studies that find an effect are more likely to be published and publicized than studies that find nothing, especially when a company with a product to sell paid for them.

Five small studies, one headline



A company that funds five small studies and promotes the one that came out well has not lied about that study. It has simply let the public record tell a story that all five together might not.

Bottom line

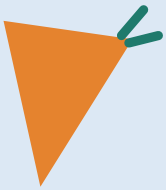
"Science-backed" is not a scientific category. What matters is the quality of the evidence behind a claim.

A cautionary tale: beta-carotene

10

In the 1980s, researchers reasoned: beta-carotene is an antioxidant, smoke causes oxidative damage, and people who eat lots of beta-carotene-rich vegetables have less lung cancer. So giving smokers beta-carotene should protect them. Every step was reasonable. The idea was so appealing that supplements were encouraged before the big trials were finished.

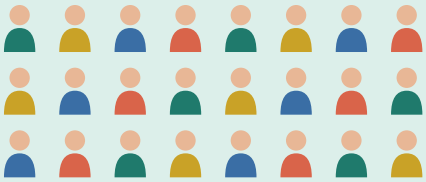
1 The idea



People who eat beta-carotene-rich veggies get less lung cancer.

So: give smokers the pill?

2 The trials



29,000+ male smokers in Finland, randomly assigned: pill or placebo

3 The surprise

+18%

more lung cancer

Also 8% more deaths. A second big trial pointed the same way.

4 The correction

Recommendation
CHANGED

Science did its job: the test beat the theory.

Two large randomized trials tested it. One enrolled more than 29,000 male smokers in Finland. The other enrolled smokers, former smokers and people exposed to asbestos in the United States. The results were the opposite of what everyone expected.

- In the Finnish trial, people taking beta-carotene had **18 percent more** lung cancer diagnoses and **8 percent more** deaths than people on a placebo.
- The U.S. trial found a similar, serious increase.
- The U.S. trial was stopped nearly two years early rather than keep exposing people to what now looked like harm.

Notice the order. The theory did not predict the result. The result came first, and scientists then had to revise the theory to explain it. A biological explanation that sounds airtight is not the same as a result you have actually observed.

How science corrects itself

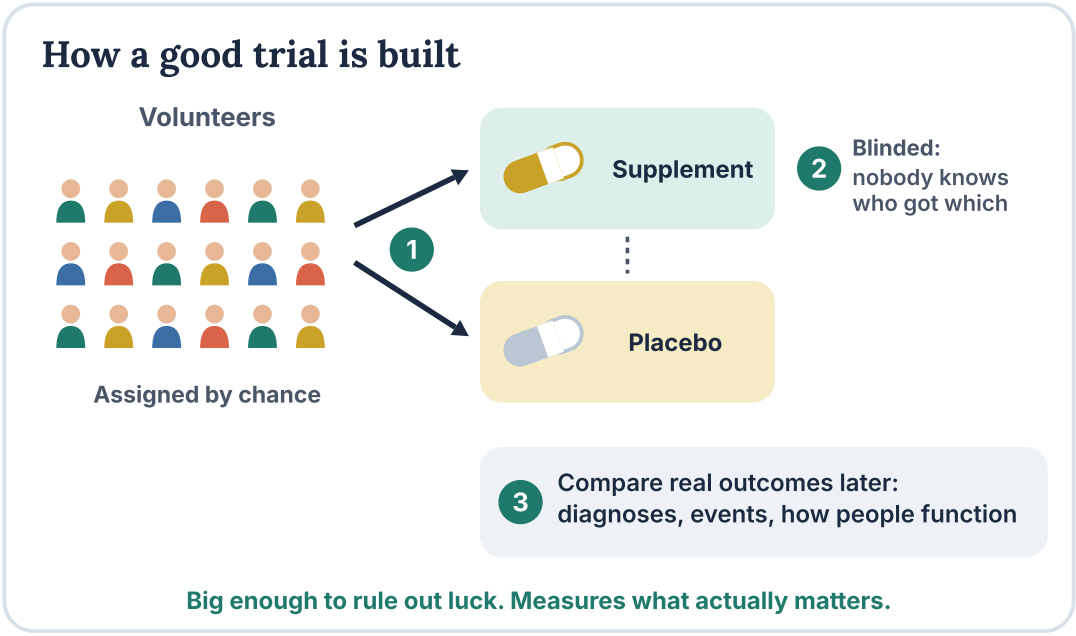


Two lessons

1. Plausible does not mean proven. **2.** The system worked: a big, well-designed trial changed the advice. The worry is how many products on shelves today have never faced a test like it.

11 What a good trial looks like

Go back to the Finnish trial and look at how it was built. Almost every feature of a trustworthy trial is in that one story.



- **Randomization.** Chance, not choice, decides who gets what, so the two groups are alike, on average, in every way except the pill.
- **Blinding.** Participants, and ideally the people measuring results, do not know who got what. This closes the door on the placebo effect and unconscious bias.
- **Size and endpoints.** That trial studied tens of thousands of people and counted a hard outcome: a confirmed cancer diagnosis. Many studies behind supplement ads involve a few dozen people for a few weeks and measure a lab value or a self-rated score.

A number that can mislead

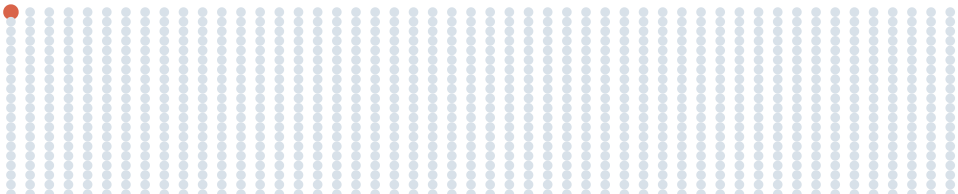
Say a supplement "cuts your risk in half." If the condition affects 2 people in 1,000 without it, then with it, about 1 in 1,000. That is a real effect, but it is one person in a thousand, not the dramatic change "in half" suggests.

"Cuts your risk in half!" Let's look closer.

Without the supplement: 1,000 people, 2 get the condition



With the supplement: 1,000 people, 1 gets the condition



Relative risk: cut in half (50%). Absolute risk: 1 person in 1,000 (0.1%).
Both are true. Only one makes the label.



Translate the claim

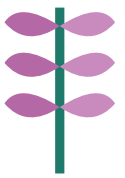
"Clinically proven" can mean a small, short study with a self-rated score and no placebo group. "Backed by science" can mean one favorable, company-funded study. "Boosts immunity" can mean one lab marker moved, with no evidence of fewer illnesses.

"Natural" does not mean safe

12

Foxglove is a wildflower. It grows in gardens across the country, entirely natural. It is also the original source of digitalis, a heart drug so powerful that, at the wrong dose, it can stop a heart. What makes digitalis usable as a medicine is precise dosing, careful monitoring and decades of study. A "supports heart health" supplement claim needs none of that.

All natural. All dangerous in the wrong hands.



Foxglove

Source of a heart drug. Too much can stop a heart.



Ephedra

Banned from supplements in 2004 after deaths.



Comfrey

Can badly injure the liver.

Safety was never a matter of where a compound comes from. It depends on how toxic it is, how much you take, and how it interacts with a particular person's body. That is as true of a plant extract as of a lab-made chemical.

Natural, and dangerous

- **Ephedra** was sold for weight loss and athletic performance. It was linked to heart attacks, strokes and sudden deaths. The FDA banned it from supplements in 2004, the only time the agency has used its supplement-law safety authority to remove an entire ingredient category.
- **Comfrey** has centuries of traditional use, yet contains compounds that can cause serious, sometimes irreversible liver injury. Many countries restrict or ban it for internal use.
- **Aristolochic acid**, found in some traditional botanical products, has been linked to kidney failure and some urinary tract cancers.



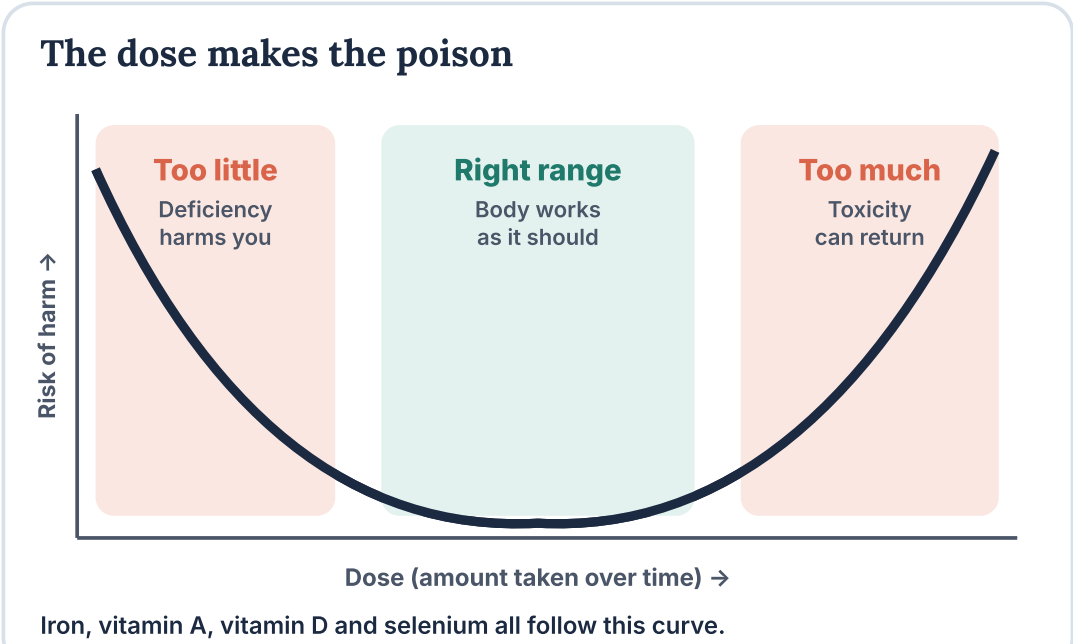


Who is most at risk

Pregnant and breastfeeding women, older adults taking several medicines, people with liver or kidney problems, and children. Very few supplements have been studied at specific doses for any of these groups.

13 The dose makes the poison

Iron is essential. Without enough, you become anemic and tired. Iron overdose is also a well-known cause of serious, sometimes fatal poisoning in young children, because a handful of adult iron tablets can be a toxic dose for a small child. The same substance sits on both sides of that sentence, separated only by dose.

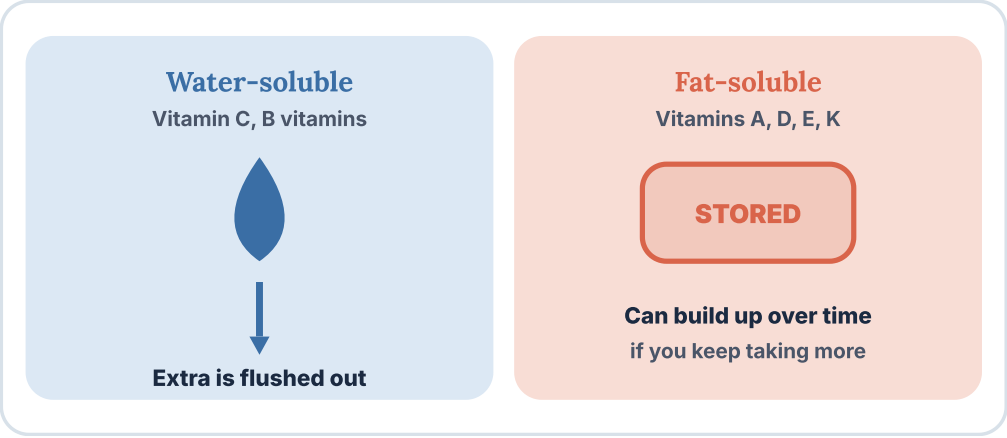


Iron, vitamin A, vitamin D and selenium all follow this curve.

Every essential nutrient has a recommended intake, and many have a **Tolerable Upper Intake Level**, a ceiling above which the risk of harm rises. If more were always better, nutrition science would have no reason to set a ceiling at all.

Why some vitamins need extra care

Water-soluble vitamins (C and the B vitamins) are mostly flushed out in urine when you take more than you need. Fat-soluble vitamins (A, D, E and K) are stored in the body and can build up to toxic levels over time.



- **Vitamin A** in high doses can damage the liver, and in pregnancy can cause serious birth defects.
- **Vitamin D** at high doses over time can raise blood calcium enough to affect the heart and kidneys.
- **Selenium**, an essential mineral, can cause hair and nail changes and nerve damage at chronic high intakes.

Bottom line

"More support" assumes more is always better. For many of the nutrients in supplements, the risk of harm climbs again once the dose gets too high.

14 Supplements and your medicines

In 2000, doctors reported a worrying pattern in transplant patients on cyclosporine, a drug that prevents organ rejection. Their drug levels were dropping dangerously, and some had acute rejection. The cause was an ordinary, legal, over-the-counter herb that some had started for mood: St. John's wort.

Supplement + medicine = possible trouble

St. John's wort	+	Transplant, birth control, some cancer drugs	→ Medicine works less
Fish oil, ginkgo, high-dose vitamin E	+	Blood thinner (warfarin)	→ More bleeding risk
Chromium, cinnamon, bitter melon	+	Diabetes medicine	→ Blood sugar too low
Melatonin, valerian, kava	+	Sleep or anxiety medicine	→ Too sleepy

Tell your doctor and pharmacist about everything you take.

Supplements can interact with medicines the same way medicines interact with each other: by speeding up or slowing the liver's breakdown of a drug, by adding to its effect, or by changing absorption. St. John's wort also lowers the effect of birth control pills and some cancer drugs.

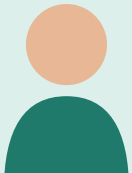
Why your doctor needs the full list

Supplements are a recognized cause of liver injury, and the label may not tell you everything that is in the bottle. In a registry of serious liver injuries funded by the National Institutes of Health (NIH), the share blamed on herbal and dietary supplements rose from 7 percent to 20 percent between 2004 and 2013. In one study of liver injury linked to green tea extract, 29 of 73 implicated products (40 percent) did not list green tea extract on the label at all. The registry tracks referred cases, so it does not give an overall rate. The practical lesson comes in three parts: doctors often do not ask, you need to bring your supplements, and you should tell everyone who treats you.



Doctors do not always ask

A common story: a patient has unexplained liver tests, "any medications?" turns up nothing, and only the direct question "any vitamins, herbs or supplements?" finds the cause.



Bring the whole bag to your next visit.
Prescription bottles, vitamins, herbs, powders.



Bottom line

Tell every doctor and pharmacist about everything you take. Do not rely on the label or on a lack of warnings.

The report card

15

Here is how six common supplement claims hold up when the best large trials are examined. It is not a list of things to take or avoid, and it is not personal medical advice. It shows how wide the gap can be between a label and a result.

The report card: claim versus evidence

Omega-3 (fish oil) Heart health	Partly	Fewer heart attacks in one big trial, but no overall drop in major heart events
Vitamin D, omega-3 Cancer prevention	Not supported	No significant reduction in cancers
Ginkgo Memory, dementia	Not supported	3,000+ older adults, 6 years: no protection
Chromium, cinnamon Blood sugar	Partly	Small, inconsistent effects; no proven disease benefit
Calcium + vitamin D Fewer fractures	Supported	Benefit in older adults with low intake; some heart questions remain
Vitamin B12 Energy	Partly	Helps if you are deficient; unproven if you are not

Supported · Partly, or only in some people · Not supported

How to read this card

"Supported" means a big trial, or repeated trials, found a benefit for that specific claim. "Not supported" means the best trial found no benefit. It does not mean the supplement is harmful.

A closer look

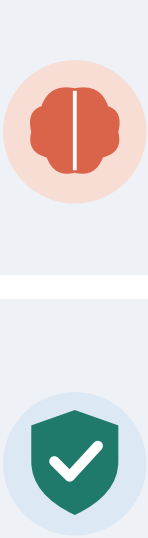


Fish oil and the heart

In a trial of nearly 26,000 adults, omega-3 did not significantly reduce the main combined measure of major heart events. Inside it, though, heart attacks fell 28 percent and fatal heart attacks 50 percent, with no benefit for stroke. A narrower benefit than the label implies, but not nothing. Those were secondary results, so they are less certain than the main finding.

Ginkgo and memory

One of the largest trials of this question followed more than 3,000 older adults for about six years. 277 people on ginkgo developed dementia versus 246 on placebo. No protection.



Calcium and vitamin D

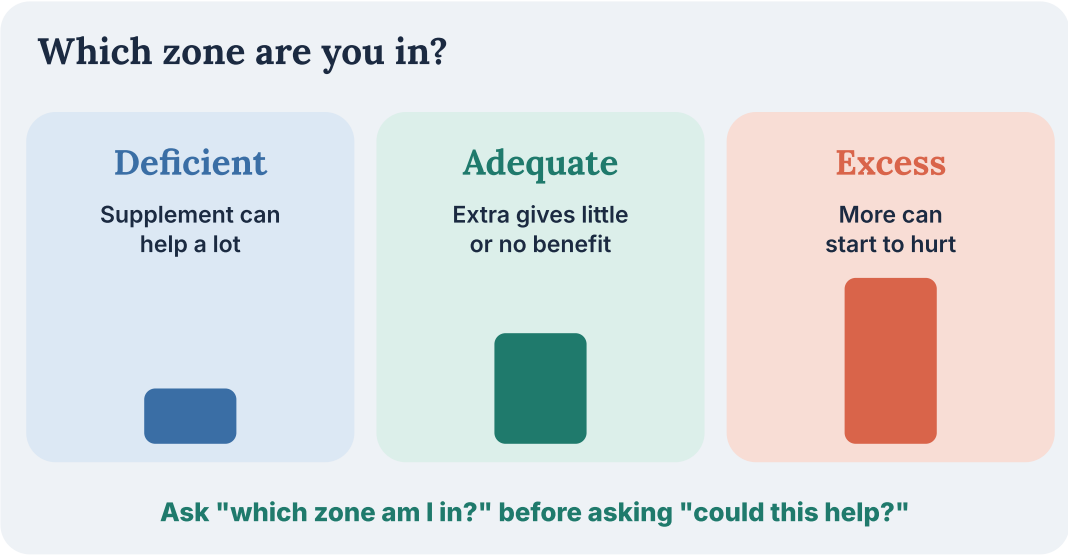
A rare success: in older adults with low intake, especially in care homes, trials show fewer fractures. Results in healthy people living at home are less consistent. Some analyses have raised a modest heart-risk question about calcium, which is still being studied.

In the same large trial, vitamin D and omega-3 did not significantly lower cancer or major heart events. Chromium and cinnamon have small, inconsistent effects on blood sugar and no proven disease benefit.

Do I need one? Three zones

16

A vitamin's value depends mostly on something the label cannot see: whether you are actually short of it. Think of three zones instead of two.



Someone who is truly deficient can see a real, sometimes dramatic improvement from correcting it. Someone who already has enough sees, at best, no added benefit, and past a point can move into the third zone.

Ask this first

Before asking "could this help me?" ask "which of the three zones am I actually in?" Only a doctor, with the right test or history, can tell you.

Three everyday examples



Vitamin C
The body flushes out extra, so serious overdose is rare. That apparent safety helps explain why megadoses are sold. But decades of trials show little or no effect on colds in the general public, and no cancer benefit.



Vitamin D
Strong evidence for bone and muscle health in people who are truly deficient. In large trials, no significant effect on cancer or major heart problems.



B12 and "energy"
B12 treats a real deficiency, which is common in older adults, some medicine users and vegans, and the improvement can feel dramatic. That does not mean it adds energy for someone whose B12 was already normal.



Bottom line
Correcting a real shortage can help. Adding more of what you already have rarely adds that benefit. It just moves you along the curve.

Your five-question checklist

17

You now have the main tools. Use the same five questions before you buy anything. They are the same ones used in the full investigation on the website, so you can check your own answers against it.

Your five-question checklist

- [ ] 1 **What does it cost?**
  Per serving and per year, including how to cancel a subscription.
- [ ] 2 **What is promised?**
  The exact words on the label and in the ads.
- [ ] 3 **What has been studied?**
  Real trials, how big, how long, who paid, what outcome.
- [ ] 4 **What is still unknown?**
  Hidden amounts, long-term safety, medicine interactions.
- [ ] 5 **What is the track record?**
  Warnings, lawsuits, recalls. Check the dates.

Still unsure? Ask your pharmacist. It is free.

Keep it simple

1. **Cost:** work out the price per serving and per year. Find out how to cancel a subscription before you sign up.
2. **Promise:** write down the exact claim. Is it a disease claim or a "supports" claim?
3. **Studied:** look for real trials, not testimonials. How big, how long, who paid, what was measured?
4. **Unknown:** any hidden blends, unstudied combinations, or medicine interactions?
5. **Track record:** any warnings, lawsuits or recalls? Always check the date.

If the answer to any question is "I cannot tell," that is itself useful information. And remember: your pharmacist is a free, trained second opinion, especially on medicine interactions.



Plain-English glossary

Adverse event. Any bad reaction reported after using a product.

Blinding. Keeping people in a study unaware of who got the real treatment and who got a placebo.

CAERS. The FDA's reporting system for bad reactions tied to foods and supplements.

cGMPs. Current good manufacturing practices: FDA rules for how supplements are made, tested and labeled.

Deficiency. Not having enough of a nutrient. The case where supplements are most likely to help.

Disease claim. A claim to treat, cure or prevent a disease. Not allowed on a supplement.

Dose. The amount of something you take, and how often.

DSHEA. The 1994 law that set today's rules for U.S. dietary supplements.

Evidence ladder. A ranking of study types from weakest to strongest.

FDA. The U.S. Food and Drug Administration, the federal agency that regulates drugs, foods and supplements.

Placebo. A look-alike with no active ingredient, used for comparison in a trial.

Proprietary blend. A mix whose total weight is listed but whose individual amounts are secret.

Randomized controlled trial. A study that assigns people by chance to a treatment or a comparison.

Structure/function claim. A claim that a product "supports" a normal body function. Needs no FDA approval.

Tolerable Upper Intake Level. The highest daily amount unlikely to cause harm for most people.

Where these facts come from

Every fact in this primer comes from the three-book investigation. Each book lists its sources, so you can check any claim yourself.

- **Book One: What the Label Doesn't Tell You.** Chapters 1 to 7: the supplement market, marketing, anecdote, the evidence ladder, mechanism versus proof, clinical trials, and beta-carotene.
- **Book Two: The Hollow Aisle.** Chapters 1 to 8: the law, the label, claims, post-sale oversight, natural versus safe, documented harms, interactions, dose.
- **Book Three: Prove It.** Chapters 1, 2, 9 and 10, and the appendix: evidence on common claims, vitamins, and the ephedra case.

Read the full books, the evidence library and the product lookup free at **supplements.brokenpromiseshealthcare.org**. The SuppleFacts app helps you check a real product label before you buy.

A note on how we write

We say when evidence is thin, when a result comes from a single study, and when something is still being studied. If you believe something here is wrong, tell us.



FREE · KEEP GOING

Your next two stops

This primer is the starting line. Both of these are free, and both are a scan away.



GO DEEPER

The American Supplement Investigation

Three free books with sources for every claim, plus the evidence library and the Product Lookup.

supplements.brokenpromiseshealthcare.org



USE IT AT THE SHELF

The SuppleFacts app

Scan a bottle, see its real label from the NIH database, and check each ingredient before you buy.

supplefacts.brokenpromiseshealthcare.org

Point your phone camera at a code. No app-store download, no sign-up.



Follow the money.
Follow the science.
Follow the evidence.

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